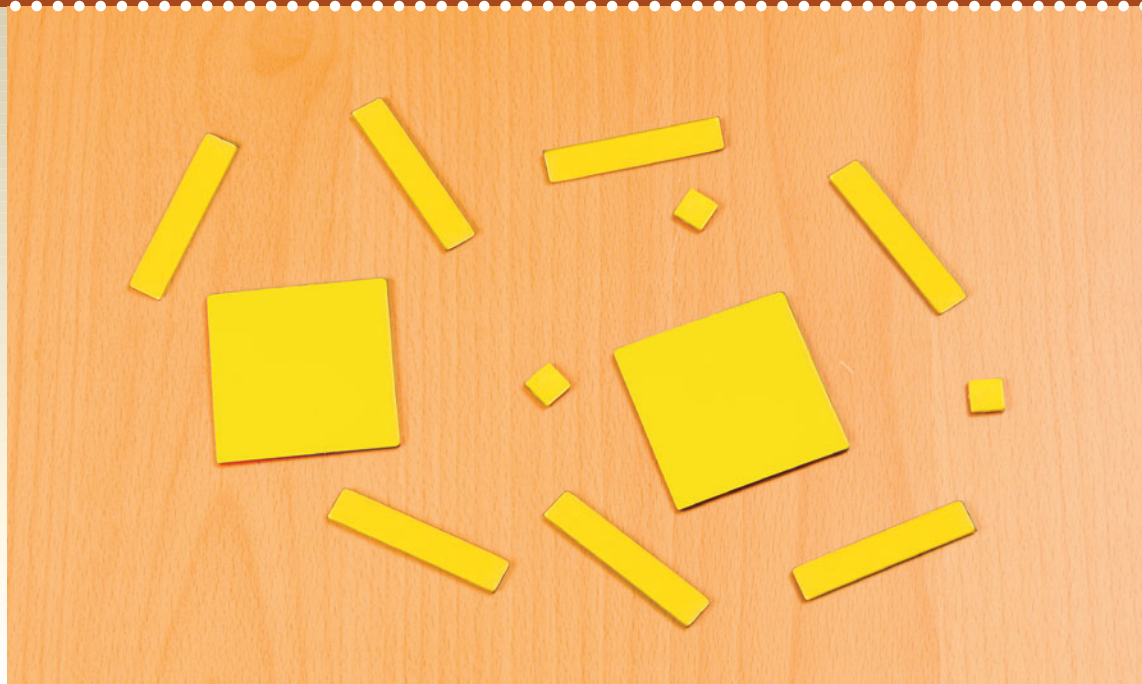


3.6 Polynomials of the Form $ax^2 + bx + c$

LESSON FOCUS

Extend the strategies for multiplying binomials and factoring trinomials.



Make Connections

Which trinomial is represented by the algebra tiles shown above?

How can the tiles be arranged to form a rectangle?

Construct Understanding

THINK ABOUT IT

Work with a partner.

You will need algebra tiles.

For which of these trinomials can the algebra tiles be arranged to form a rectangle? For those that can be represented by a rectangular arrangement, write the multiplication sentence.

$$2x^2 + 15x + 7$$

$$2x^2 + 5x + 2$$

$$6x^2 + 7x + 2$$

$$5x^2 + 4x + 4$$

$$2x^2 + 9x + 10$$

$$5x^2 + 11x + 2$$

To multiply two binomials where the coefficients of the variables are not 1, we use the same strategies as for the binomials in Lesson 3.5.

Example 1**Multiplying Two Binomials with Positive Terms**

Expand: $(3d + 4)(4d + 2)$

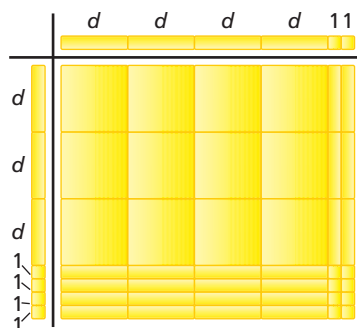
SOLUTIONS**Method 1**

Use algebra tiles.

Make a rectangle with dimensions $3d + 4$ and $4d + 2$.

Place tiles to represent each dimension, then fill the rectangle with tiles.

The tiles that form the product represent $12d^2 + 22d + 8$.



$$\text{So, } (3d + 4)(4d + 2) = 12d^2 + 22d + 8$$

Method 2

Use an area model.

To expand: $(3d + 4)(4d + 2)$, draw a rectangle with dimensions $3d + 4$ and $4d + 2$. Divide the rectangle into 4 smaller rectangles then calculate the area of each.

	$3d$	4
$4d$	$(4d)(3d) = 12d^2$	$(4d)(4) = 16d$
2	$(2)(3d) = 6d$	$(2)(4) = 8$

The area of the larger rectangle is the sum of the areas of the smaller rectangles.

$$\begin{aligned}\text{So, } (3d + 4)(4d + 2) &= 12d^2 + 16d + 6d + 8 \\ &= 12d^2 + 22d + 8\end{aligned}$$

CHECK YOUR UNDERSTANDING

1. Expand: $(5e + 3)(2e + 4)$

[Answer: $10e^2 + 26e + 12$]

This rectangle diagram shows the relationship between the coefficients in the product and the coefficients in the factors for $(ax + b)(cx + d)$.

$$(ax + b)(cx + d) = acx^2 + adx + bcx + bd$$

	cx	d
ax	$(ax)(cx) = acx^2$	$(ax)(d) = adx$
b	$(b)(cx) = bcx$	$(b)(d) = bd$

When binomials contain negative terms, it can be difficult to model their product with algebra tiles. We can use the distributive property to determine their product.

Example 2

Multiplying Two Binomials with Negative Coefficients

Expand and simplify: $(-2g + 8)(7 - 3g)$

SOLUTIONS

Method 1

Use a rectangle diagram.

Write $(-2g + 8)(7 - 3g)$ as $[(-2g) + 8][7 + (-3g)]$.

Draw a rectangle with dimensions $(-2g) + 8$ and $7 + (-3g)$.

Divide the rectangle into 4 smaller rectangles and label each one.

	7	$-3g$
$-2g$	$(-2g)(7) = -14g$	$(-2g)(-3g) = 6g^2$
8	$(8)(7) = 56$	$(8)(-3g) = -24g$

$$\begin{aligned} (-2g + 8)(7 - 3g) &= -14g + 6g^2 + 56 - 24g \\ &= 6g^2 - 14g - 24g + 56 \\ &= 6g^2 - 38g + 56 \end{aligned}$$

Method 2

Use the distributive property.

$$\begin{aligned} (-2g + 8)(7 - 3g) &= (-2g)(7 - 3g) + 8(7 - 3g) \\ &= -14g + 6g^2 + 56 - 24g \\ &= -38g + 6g^2 + 56 \\ &= 6g^2 - 38g + 56 \end{aligned}$$

CHECK YOUR UNDERSTANDING

2. Expand and simplify:

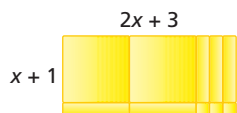
$$(6t - 9)(7 - 5t)$$

[Answer: $-30t^2 + 87t - 63$]

Which terms of the binomial factors determine the value of each term in the trinomial? How can you use this to simplify the calculation?

We can use algebra tiles to factor a trinomial when its terms are positive.

For example, to factor $2x^2 + 5x + 3$, arrange the tiles that represent these terms in a rectangle.



This algebra-tile model illustrates how the terms in the two binomial factors are related to the terms in the trinomial.

$$2x^2 + 5x + 3 = (2x + 3)(x + 1)$$

$2x^2$ is the product of the 1st terms in the binomials.

$$2x^2 + 5x + 3 = (2x + 3)(x + 1)$$

$5x$ is the sum of these products: $2x(1) + 3(x)$

$$2x^2 + 5x + 3 = (2x + 3)(x + 1)$$

3 is the product of the 2nd terms in the binomials.

When we use algebra tiles to factor a trinomial with negative terms, we may have to use the Zero Principle and add pairs of opposite tiles to form a rectangle. For example, to model the factoring of the trinomial $3x^2 + 5x - 2$, first arrange the tiles to attempt to form a rectangle. To complete the rectangle, we need two x -tiles. So, add one positive x -tile and one negative x -tile. Place them so that all the positive x -tiles are together.



$$\text{So, } 3x^2 + 5x - 2 = (3x - 1)(x + 2)$$

Some trinomials, for example $2x^2 + x + 1$, cannot be factored. You cannot add pairs of opposite x -tiles to form a rectangle.

What do you know about a polynomial for which you can never form a rectangle?

We can use the patterns on page 171 and logical reasoning to factor without algebra tiles. Often, by looking carefully at the relative sizes of the coefficients, we can estimate which factors of the coefficients of the trinomial to try first.

Example 3

Factoring a Trinomial Using Logical Reasoning

Factor.

a) $4h^2 + 20h + 9$ b) $6k^2 - 11k - 35$

SOLUTIONS

a) Method 1

$$4h^2 + 20h + 9$$

To factor this trinomial, find factors of the form $(ah + b)(ch + d)$.

The coefficient of h^2 is 4, so the coefficients of the 1st terms in the binomial are factors of 4, which are 1 and 4, or 2 and 2.

So, the binomials have the form

$$(h + b)(4h + d) \quad \text{or} \quad (2h + b)(2h + d)$$

The constant term in the trinomial is 9, so the 2nd terms in the binomial are factors of 9, which are 1 and 9, or 3 and 3.

So, the binomials could be:

$$(h + 1)(4h + 9) \quad \text{or} \quad (2h + 1)(2h + 9) \quad \text{or}$$

$$(h + 9)(4h + 1) \quad \text{or} \quad (2h + 9)(2h + 1) \quad \text{or}$$

$$(h + 3)(4h + 3) \quad \text{or} \quad (2h + 3)(2h + 3)$$

Check which of the 6 binomial products above has its h -term equal to $20h$.

$$(h + 1)(4h + 9) = 4h^2 + 13h + 9$$

$$(2h + 1)(2h + 9) = 4h^2 + 20h + 9$$

This is the correct trinomial so we need not check any further.

$$4h^2 + 20h + 9 = (2h + 1)(2h + 9)$$

(Solution continues.)

CHECK YOUR UNDERSTANDING

3. Factor.

a) $4g^2 + 11g + 6$

b) $6m^2 - 7m - 10$

[Answers: a) $(4g + 3)(g + 2)$

b) $(6m + 5)(m - 2)$]

In *Example 3a*, why did we not need to list or check the negative factors of 4?

Method 2

$$4h^2 + 20h + 9$$

Use number sense and reasoning, with mental math.

The factors of $4h^2$ are $4h$ and $1h$, or $2h$ and $2h$.

The factors of 9 are 9 and 1, or 3 and 3.

The coefficient of the middle term of the trinomial, 20, is greater than the coefficients of the other two terms, so try the greater factors. Arrange the factor combinations vertically.

$$\begin{array}{r} 4h \quad 9 \\ \times \quad \times \\ 1h \quad 1 \end{array}$$

$$\begin{array}{r} 2h \quad 9 \\ \times \quad \times \\ 2h \quad 1 \end{array}$$

Form the products,
then add.

$$4h + 9h = 13h$$

$$2h + 18h = 20h$$

← This is the correct
combination of factors.

$$\text{So, } 4h^2 + 20h + 9 = (2h + 9)(2h + 1)$$

b) Method 1

$$6k^2 - 11k - 35$$

Find factors of the form $(ak + b)(ck + d)$.

The coefficient of k^2 is 6, so the coefficients of the 1st terms in the binomial are factors of 6, which are 1 and 6, or 2 and 3.

The constant term in the trinomial is -35 , so the 2nd terms in the binomial are factors of -35 , which are 1 and -35 , or -1 and 35, or -5 and 7, or 5 and -7 .

The coefficient of the middle term of the trinomial, -11 , is between the coefficients of the other two terms, so try combinations of the lesser factors first; that is, try combinations of 2 and 3 with combinations of ± 5 and ± 7 .

So, the binomials could be:

$$\begin{array}{lll} (2k + 5)(3k - 7) & \text{or} & (2k - 7)(3k + 5) \\ (2k - 5)(3k + 7) & \text{or} & (2k + 7)(3k - 5) \end{array}$$

Check if any of the 4 binomial products above has its k -term equal to $-11k$.

$$(2k + 5)(3k - 7) = 6k^2 + k - 35$$

$$(2k - 7)(3k + 5) = 6k^2 - 11k - 35$$

This is the correct trinomial so we need not check any further.

$$6k^2 - 11k - 35 = (2k - 7)(3k + 5)$$

(Solution continues.)

In Example 3b, why did we not need to list or check the negative factors of 6?

Method 2

$$6k^2 - 11k - 35$$

Use number sense and mental math, with guess and test.

The factors of $6k^2$ are $1k$ and $6k$, or $2k$ and $3k$.

The factors of -35 are 1 and -35 , or -1 and 35 , or -7 and 5 , or 7 and -5 .

The coefficient of the middle term of the trinomial, -11 , is between the coefficients of the other terms, so try factors of these coefficients that would be in the middle if the factors were listed in numerical order. Arrange the factor combinations vertically:

$$\begin{array}{r} 2k \quad 7 \\ \times \\ 3k \quad -5 \end{array}$$

$$-10k + 21k = 11k$$

$$6k^2 - 11k - 35 = (2k - 7)(3k + 5)$$

$$\begin{array}{r} 2k \quad -7 \\ \times \\ 3k \quad 5 \end{array}$$

$$10k - 21k = -11k$$

Form the products, then add.

← This is the correct combination of factors.

Factoring by decomposition

is factoring after writing the middle term of a trinomial as a sum of two terms, then determining a common binomial factor from the two pairs of terms formed.

Another method of factoring is **factoring by decomposition**.

Consider the binomial product: $(3h + 4)(2h + 1)$

We can use the distributive property to expand:

$$\begin{aligned} (3h + 4)(2h + 1) &= 3h(2h + 1) + 4(2h + 1) \\ &= 6h^2 + 3h + 8h + 4 \\ &= 6h^2 + 11h + 4 \end{aligned}$$

To factor $6h^2 + 11h + 4$ by decomposition, we reverse the steps above.

Notice that the coefficients of the h -terms have the product: $3(8) = 24$

This is equal to the product of the coefficient of the h^2 -term and the constant term: $6(4) = 24$

So, to factor $6h^2 + 11h + 4$, we *decompose* the h -term and write it as a sum of two terms whose coefficients have a product of 24.

Why don't we need to list the negative factors of 24?

Factors of 24	Sum of Factors
1, 24	$1 + 24 = 25$
2, 12	$2 + 12 = 14$
3, 8	$3 + 8 = 11$
4, 6	$4 + 6 = 10$

The two coefficients that have a sum of 11 are 3 and 8, so we write the trinomial $6h^2 + 11h + 4$ as $6h^2 + 3h + 8h + 4$.

We remove a common factor from the 1st pair of terms, and from the 2nd pair of terms:

$$6h^2 + 3h + 8h + 4 = 3h(2h + 1) + 4(2h + 1)$$

Each product has the common binomial factor $2h + 1$.

$$6h^2 + 11h + 4 = (2h + 1)(3h + 4)$$

Example 4**Factoring a Trinomial by Decomposition**

Factor.

a) $3s^2 - 13s - 10$

b) $6x^2 - 21x + 9$

SOLUTION

a) $3s^2 - 13s - 10$

Check for common factors; there are none.

The product of the coefficient of s^2 and the constant term is: $3(-10) = -30$

Write $-13s$ as the sum of two terms whose coefficients have a product of -30 .

Factors of -30	Sum of Factors
1, -30	$1 - 30 = -29$
$-1, 30$	$-1 + 30 = 29$
2, -15	$2 - 15 = -13$
$-2, 15$	$-2 + 15 = 13$
3, -10	$3 - 10 = -7$
$-3, 10$	$-3 + 10 = 7$
5, -6	$5 - 6 = -1$
$-5, 6$	$-5 + 6 = 1$

The two coefficients are 2 and -15 , so write the trinomial $3s^2 - 13s - 10$ as $3s^2 + 2s - 15s - 10$.

Remove a common factor from the 1st pair of terms, and from the 2nd pair of terms:

$$3s^2 + 2s - 15s - 10 = s(3s + 2) - 5(3s + 2)$$

Each product has a common binomial factor.

$$3s^2 + 2s - 15s - 10 = (3s + 2)(s - 5)$$

$$\text{So, } 3s^2 - 13s - 10 = (3s + 2)(s - 5)$$

b) $6x^2 - 21x + 9$

Check for common factors; 3 is a common factor.

$$\text{So, } 6x^2 - 21x + 9 = 3(2x^2 - 7x + 3)$$

Factor $2x^2 - 7x + 3$.

(Solution continues.)

CHECK YOUR UNDERSTANDING

4. Factor.

a) $8p^2 - 18p - 5$

b) $24h^2 - 20h - 24$

[Answers: a) $(2p - 5)(4p + 1)$

b) $4(2h - 3)(3h + 2)$]

When we write the middle term as the sum of two terms, could we write $3s^2 - 15s + 2s - 10$ instead? Justify your answer.

The product of the coefficient of x^2 and the constant term is: $2(3) = 6$

Write $-7x$ as the sum of two terms whose coefficients have a product of 6.

Factors of 6	Sum of Factors
1, 6	$1 + 6 = 7$
-1, -6	$-1 - 6 = -7$
2, 3	$2 + 3 = 5$
-2, -3	$-2 - 3 = -5$

The two coefficients are -1 and -6 , so write the trinomial $2x^2 - 7x + 3$ as $2x^2 - 1x - 6x + 3$.

Remove a common factor from the 1st pair of terms, and from the 2nd pair of terms:

$$2x^2 - 1x - 6x + 3 = x(2x - 1) - 3(2x - 1)$$

Each product has a common binomial factor.

$$2x^2 - 1x - 6x + 3 = (2x - 1)(x - 3)$$

$$\text{So, } 2x^2 - 7x + 3 = (2x - 1)(x - 3)$$

$$\text{And, } 6x^2 - 21x + 9 = 3(2x - 1)(x - 3)$$

When would you use decomposition to factor a trinomial?

To check that the factors are correct, multiply them.

In *Example 4b*, multiply:

$$\begin{aligned} 3(2x - 1)(x - 3) &= 3(2x^2 - 6x - x + 3) \\ &= 3(2x^2 - 7x + 3) \\ &= 6x^2 - 21x + 9 \end{aligned}$$

Since this trinomial is the same as the original trinomial, the factors are correct.

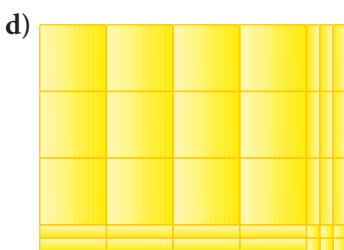
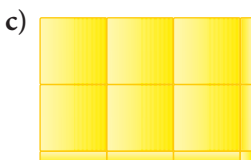
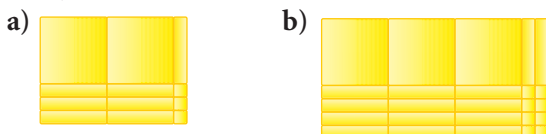
Discuss the Ideas

1. How are the strategies for multiplying binomial factors of the form $(ax + b)(cx + d)$ like the strategies you used in Lesson 3.5 for products of the form $(x + a)(x + b)$? How are they different?
2. How can you use the coefficients of a trinomial to determine the coefficients of its binomial factors?
3. How can you tell when a trinomial cannot be factored?
4. How can you use logical reasoning to reduce the number of possible combinations of coefficients you have to consider in the guess-and-test or the decomposition methods of factoring a trinomial?

Exercises

A

5. Write the multiplication sentence that each set of algebra tiles represents.

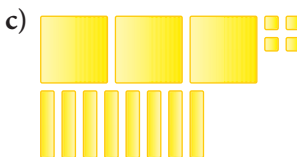
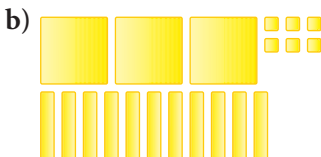


6. Use algebra tiles to determine each product.

- a) $(2v + 3)(v + 2)$ b) $(3r + 1)(r + 4)$
 c) $(2g + 3)(3g + 2)$ d) $(4z + 3)(2z + 5)$
 e) $(3t + 4)(3t + 4)$ f) $(2r + 3)(2r + 3)$

7. For each set of algebra tiles below:

- i) Write the trinomial that the algebra tiles represent.
 ii) Arrange the tiles to form a rectangle.
 Sketch the rectangle.
 iii) Use the rectangle to factor the trinomial.



B

8. Copy and complete each statement.

- a) $(2w + 1)(w + 6) = 2w^2 + \square w + 6$
 b) $(2g - 5)(3g - 3) = 6g^2 + \square + \square$
 c) $(-4v - 3)(-2v - 7) = \square + \square + 21$

9. Expand and simplify.

- a) $(5 + f)(3 + 4f)$
 b) $(3 - 4t)(5 - 3t)$
 c) $(10 - r)(9 + 2r)$
 d) $(-6 + 2m)(-6 + 2m)$
 e) $(-8 - 2x)(3 - 7x)$
 f) $(6 - 5n)(-6 + 5n)$

10. Expand and simplify.

- a) $(3c + 4)(5 + 2c)$
 b) $(1 - 7t)(3t + 5)$
 c) $(-4r - 7)(2 - 8r)$
 d) $(-9 - t)(-5t - 1)$
 e) $(7h + 10)(-3 + 5h)$
 f) $(7 - 6y)(6y - 7)$

11. a) Use algebra tiles to factor each polynomial.

Sketch the tiles you used.

- i) $3t^2 + 4t + 1$
 ii) $3t^2 + 8t + 4$
 iii) $3t^2 + 12t + 9$
 iv) $3t^2 + 16t + 16$

- b) What patterns do you see in the algebra-tile rectangles? How are these patterns shown in the binomial factors?
 c) Write the next 3 trinomials in the pattern and their binomial factors.

12. Factor. What patterns do you see in the trinomials and their factors?

- a) i) $2n^2 + 13n + 6$ ii) $2n^2 - 13n + 6$
 b) i) $2n^2 + 11n - 6$ ii) $2n^2 - 11n - 6$
 c) i) $2n^2 + 7n + 6$ ii) $2n^2 - 7n + 6$

13. Factor. Check by expanding.

- a) $2y^2 + 5y + 2$ b) $2a^2 + 11a + 12$
 c) $2k^2 + 13k + 15$ d) $2m^2 - 11m + 12$
 e) $2k^2 - 11k + 15$ f) $2m^2 + 15m + 7$
 g) $2g^2 + 15g + 18$ h) $2n^2 + 9n - 18$

14. a) Find two integers with the given properties.

	Product	Sum
i)	15	16
ii)	24	14
iii)	15	8
iv)	12	7
v)	12	13
vi)	24	11

- b) Use the results of part a to use decomposition to factor each trinomial.
- i) $3v^2 + 16v + 5$ ii) $3m^2 + 14m + 8$
 iii) $3b^2 + 8b + 5$ iv) $4a^2 + 7a + 3$
 v) $4d^2 + 13d + 3$ vi) $4v^2 + 11v + 6$

15. Factor. Check by expanding.

- a) $5a^2 - 7a - 6$ b) $3y^2 - 13y - 10$
 c) $5s^2 + 19s - 4$ d) $14c^2 - 19c - 3$
 e) $8a^2 + 18a - 5$ f) $8r^2 - 14r + 3$
 g) $6d^2 + d - 5$ h) $15e^2 - 7e - 2$

16. Find and correct the errors in each factorization.

- a) $6u^2 + 17u - 14 = (2u - 7)(3u + 2)$
 b) $3k^2 - k - 30 = (3k - 3)(k + 10)$
 c) $4v^2 - 21v + 20 = (4v - 4)(v + 5)$

17. Find and correct the errors in this solution of factoring by decomposition.

$$\begin{aligned} 15g^2 + 17g - 42 &= 15g^2 - 18g + 35g - 42 \\ &= 3g(5g - 6) + 7(5g + 6) \\ &= (3g + 7)(5g + 6) \end{aligned}$$

18. Factor.

- a) $20r^2 + 70r + 60$ b) $15a^2 - 65a + 20$
 c) $18h^2 + 15h - 18$ d) $24u^2 - 72u + 54$
 e) $12m^2 - 52m - 40$ f) $24g^2 - 2g - 70$

19. Factor.

- a) $14y^2 - 13y + 3$ b) $10p^2 - 17p - 6$
 c) $10r^2 - 33r - 7$ d) $15g^2 - g - 2$
 e) $4x^2 + 4x - 15$ f) $9d^2 - 24d + 16$
 g) $9t^2 + 12t + 4$ h) $40y^2 + y - 6$
 i) $24c^2 + 26c - 15$ j) $8x^2 + 14x - 15$

20. Find an integer to replace $\square$ so that each trinomial can be factored.

How many integers can you find each time?

- a) $4s^2 + \square s + 3$ b) $4h^2 + \square h + 25$
 c) $6y^2 + \square y - 9$ d) $12t^2 + \square t + 10$
 e) $9z^2 + \square z + 1$ f) $\square f^2 + 2f + \square$

C

21. a) Factor, if possible.

- i) $4r^2 - r - 5$ ii) $2t^2 + 10t + 3$
 iii) $5y^2 + 4y - 2$ iv) $2w^2 - 5w + 2$
 v) $3h^2 - 8h - 3$ vi) $2f^2 - f + 1$

- b) Choose two trinomials from part a: one that can be factored and one that cannot be factored. Explain why the first trinomial can be factored and the second one cannot be factored.

22. a) Factor each trinomial.

- i) $3n^2 + 11n + 10$ ii) $3n^2 - 11n + 10$
 iii) $3n^2 + 13n + 10$ iv) $3n^2 - 13n + 10$
 v) $3n^2 + 17n + 10$ vi) $3n^2 - 17n + 10$

- b) Look at the trinomials and their factors in part a. Are there any other trinomials that begin with $3n^2$, end with $+10$, and can be factored? Explain.

23. Find all the trinomials that begin with $9m^2$, end with $+16$, and can be factored.

Reflect

Which strategies can you use to factor a trinomial? Give an example of when you might use each strategy to factor a trinomial.



CHECKPOINT 2

Connections

Type of Factor	Patterns Relating Factors and the Product	Example
Monomial x Polynomial	The monomial factor is the greatest common factor of the terms of the product polynomial.	$5x^2 - 10xy + 15x$ $= 5x(x - 2y + 3)$
$(x + a)(x + b)$	$(x + a)(x + b)$ $= x^2 + (a + b)x + ab$	$(x + 2)(x - 8)$ $= x^2 + 2x - 8x - 16$ $= x^2 - 6x - 16$
$(ax + b)(cx + d)$	$(ax + b)(cx + d)$ $= (ac)x^2 + (ad + bc)x + bd$	$(3x - 2)(2x + 5)$ $= 6x^2 - 4x + 15x - 10$ $= 6x^2 + 11x - 10$

Concept Development

In Lesson 3.3

- You applied an area model for the product of whole numbers to develop models and strategies to determine **common factors** in the terms of a polynomial.

In Lesson 3.4

- You used algebra tiles to **multiply binomials** and to **factor trinomials**.

In Lesson 3.5

- You used diagrams and algebraic strategies to **multiply binomials** and to **factor trinomials** of the form $x^2 + bx + c$.

In Lesson 3.6

- You extended the strategies from Lesson 3.5 to **multiply binomials** and to **factor trinomials** of the form $ax^2 + bx + c$.



THE WORLD OF MATH

Careers: Computer Graphics Artist

If you have played a video game or been to a movie recently, you may have seen special effects created by a computer graphics artist. To create an image in virtual space, the artist may take measurements from a model and make many calculations to ensure the objects look and move realistically on the screen. Computer graphics artists use their knowledge of algebra and geometry to help produce visually attractive and entertaining material.



Assess Your Understanding

3.3

1. For each set of algebra tiles, write the polynomial they represent and identify its factors. Sketch the tile arrangement that illustrates the factors.



2. a) Factor each polynomial. Use algebra tiles when you can. Sketch the tiles you used.
- i) $4a + 8$
 - ii) $3c - 6$
 - iii) $-2v^2 - 5v$
 - iv) $2x^2 + 14x + 6$
 - v) $-3r^2 + 15r - 3$
 - vi) $15a^3 - 3a^2b - 6ab^2$
 - vii) $12 - 32x + 8x^2$
 - viii) $12x^2y - 8xy - 16y$
- b) For which polynomials in part a could you not use algebra tiles? Explain why you could not use them.

3.4

3. Use 1 x^2 -tile. Choose x -tiles and 1-tiles to make a rectangle. Sketch your arrangement and write the multiplication sentence it represents.
4. Use 2 or more x^2 -tiles. Choose x -tiles and 1-tiles to make a rectangle. Sketch your arrangement and write the multiplication sentence it represents.

3.5

5. Expand and simplify. Use an area model or a rectangle diagram to illustrate each product.
- a) $(x + 1)(x + 4)$
 - b) $(d - 2)(d + 3)$
 - c) $(x - 4)(x - 2)$
 - d) $(5 - r)(6 + r)$
 - e) $(g + 5)(g - 1)$
 - f) $(2 - t)(10 - t)$
6. Factor each trinomial. Check by expanding.
- a) $s^2 + 11s + 30$
 - b) $n^2 - n - 30$
 - c) $20 - 9b + b^2$
 - d) $-11 - 10t + t^2$
 - e) $z^2 + 13z + 30$
 - f) $-k^2 + 9k - 18$
7. Factor.
- a) $3x^2 + 15x - 42$
 - b) $-2y^2 + 22y - 48$
 - c) $-24 - 11m - m^2$
 - d) $50 - 23y - y^2$

3.6

8. Expand and simplify.

- | | |
|-----------------------|-------------------------|
| a) $(2c + 1)(c + 3)$ | b) $(-m + 5)(4m - 1)$ |
| c) $(3f - 4)(3f + 1)$ | d) $(6z - 1)(2z - 3)$ |
| e) $(5 - 3r)(6 + 2r)$ | f) $(-4 - 2h)(-2 - 4h)$ |

9. Factor each trinomial. Check by expanding.

- | | |
|----------------------|----------------------|
| a) $2j^2 + 13j + 20$ | b) $3v^2 + v - 10$ |
| c) $5k^2 - 23k + 12$ | d) $9h^2 + 18h + 8$ |
| e) $8y^2 - 2y - 1$ | f) $6 - 23u + 20u^2$ |



THE WORLD OF MATH

Historical Moment: François Viète

François Viète (1540–1603) was a lawyer who worked at the court of Henri III and Henri IV of France. In addition to offering legal advice, he was a cryptanalyst, deciphering messages intercepted between enemies of the king, but his hobby was mathematics. He was one of the first people to use letters to represent numbers. In his work *Prior Notes on Symbolic Logistic*, written toward the end of the sixteenth century but not published until 1631, he showed how to operate on symbolic quantities and derived many algebraic results. He applied his algebra to many different fields of mathematics, publishing work on trigonometry and what he viewed as his most elegant, the analysis of equations.

